# JAWAHARLAL NEHRU UNIVERSITY

## SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

**Mid-Term Exam**

**M.CA. (Winter Semester-2023)**

**Course Name: Computer Architecture** **Maximum Marks: 40**

**Course Code:** CS 106 **Total Time: 2.0 hours**

Q1. What is the difference between RISC and CISC type of Computer Architecture? Three enhancements with following speedups are proposed for a new computer:

$Speedup_1$=25; $Speedup_2$=15; $Speedup_3$=10; only one enhancement is useable at time

(a) If enhancement 1 and 2 are each usable for 22% of the time, what fraction of the time must enhancement 3 be used to achieve overall speed up of 15?.

(b) Assume the enhancement can be used 20%, 35% and 15% of the time for enhancement 1, 2, 3 respectively. For what fraction of the reduced execution time no enhancement is used?

(3+7)

Q2. (a) What are the major performance parameters in designing computer architecture?

b) What are various types of addressing modes in Computer Architecture?

(c) Suppose a program is running on a machine with the following instruction types, CPI values and the frequencies of occurrence. The CPU designer gives two options i) reduce CPI instruction type A to 1.0 and ii) reduce CPI instruction type B to 1.5. Which one is better?.

| Type | CPI | Frequency |
|---|---|---|
| A | 1.2 | 50% |
| B | 2.4 | 30% |
| C | 2.1 | 20% |

(3+2+5
)

Q3. (a) What is the difference between SRAM and DRAM memory cell? Where are they used in the memory hierarchy system and why?

(b) What conclusion can one draw from the following observations in terms of computing performance

| Type | CPU A (in Sec.) | CPU B (in Sec.) | CPU C (in Sec.) |
|---|---|---|---|
| Program A | 60 | 40 | 30 |
| Program B | 510 | 220 | 90 |
| Total Time | 570 | 260 | 120 |
| | | | |

Q4. (a) What is the difference between Access time and Latency in a memory system? What is the difference Associative Mapping N Set Associative Mapping?

(b) Consider a two level memory hierarchy with separate instruction and data cache in level 1 and level 2 respectively. The main memory is at level 3.
Calculate the average access time for the following parameters:

i) Clock cycle is 1 ns
ii) Miss cycle is 10 clock cycle
iii) 2 percent of instruction is not found in Instruction cache
iv) 9 percent of instruction is not found in Data cache
v) 25 percent of the total memory access for the data
vi) Cache Access Time (including hit detection) is 1 clock cycle.

(6+4)